

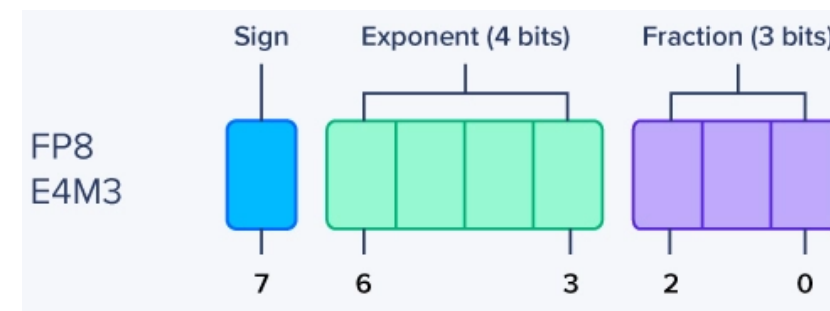
2026 DCS Lab 07 相關知識介紹

Floating Point

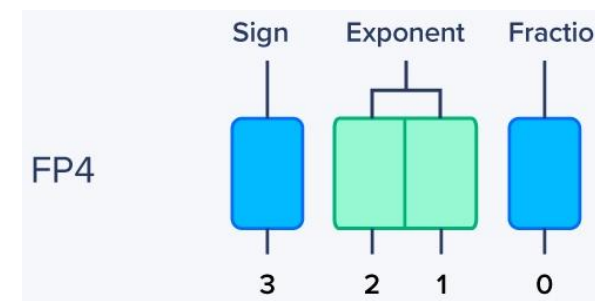
float32: IEEE single-precision



float16: IEEE half-precision



bfloat16



參考資料:

<https://medium.com/@manyi.yim/bfloat16-vs-float32-vs-float16-back-to-the-basics-80d4aec49ca8>

<https://www.exxactcorp.com/blog/hpc/what-is-fp8-fp6-fp4>

Floating Point – bfloat16

bfloat16



- Sign: 0 代表正，1 代表負
- Exponent: 2 的幾次方
- Fraction (mantissa): 小數點後的值 (1.fraction)
- 公式: $((-1)^{\text{Sign}}) * (2^{(\text{Exponent}-127)}) * (1.\text{Fraction})$

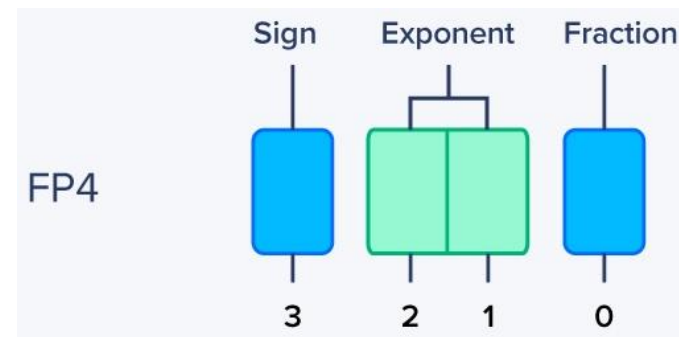
Example:

- $1 = 0\ 01111111\ 00000000 = ((-1)^0) * (2^{(127-127)}) * (1.00000000)$
- $-3 = 1\ 10000000\ 10000000 = ((-1)^1) * (2^{(128-127)}) * (1.10000000)$

- Issue: LLM(Large Language Model, e.g. ChatGPT) 中，memory 需求大，若改用 INT4、INT8 來計算，精度又會太低

Floating Point – FP4

- Sign: 0 代表正，1 代表負
- Exponent: 2 的幾次方
- Fraction (mantissa): 小數點後的值 (1.fraction)
- 公式: $((-1)^{\text{Sign}}) * (2^{(\text{Exponent}-1)}) * (1.\text{Fraction})$

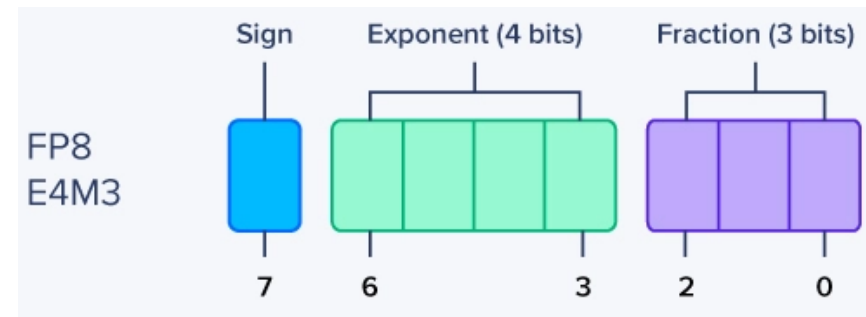


Example:

- $1 = 0\ 01\ 0 = ((-1)^0) * (2^{(1-1)}) * (1.0)$
- $-3 = 1\ 10\ 1 = ((-1)^1) * (2^{(2-1)}) * (1.1)$

Floating Point – FP8 (E4M3)

- Sign: 0 代表正，1 代表負
- Exponent: 2 的幾次方
- Fraction (mantissa): 小數點後的值 (1.fraction)
- 公式: $((-1)^{\text{Sign}}) * (2^{(\text{Exponent}-7)}) * (1.\text{Fraction})$

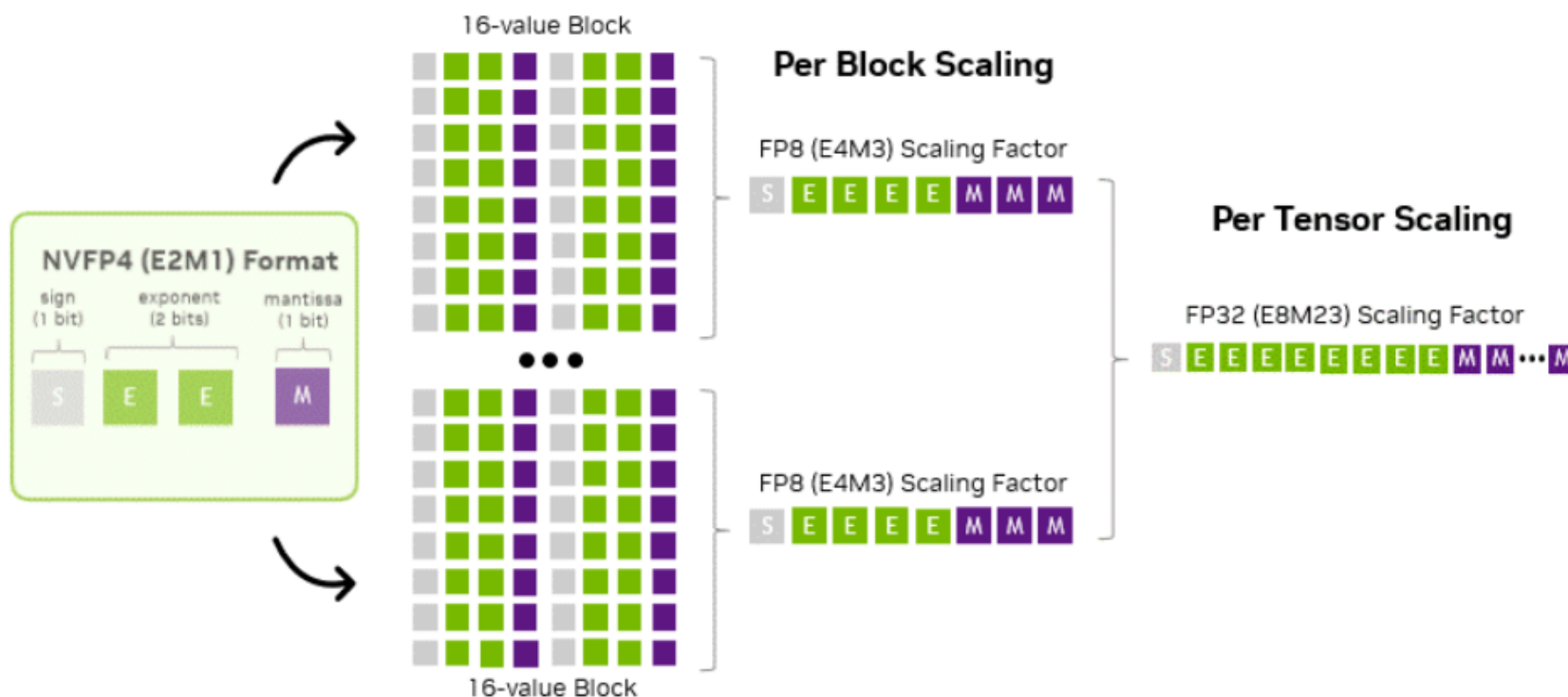


Example:

- $1 = 0\ 0111\ 000 = ((-1)^0) * (2^{(7-7)}) * (1.000)$
- $-6 = 1\ 1001\ 010 = ((-1)^1) * (2^{(9-7)}) * (1.010)$

What is NVFP4

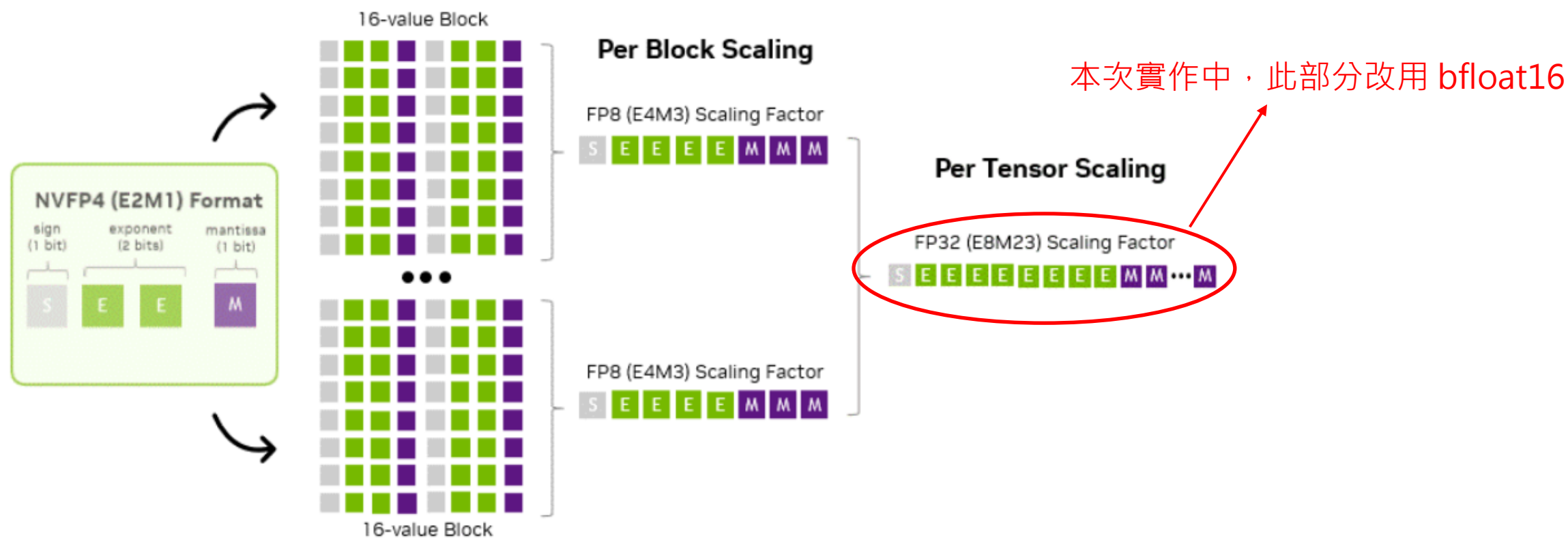
- NVIDIA 設計之 4-bit low precision 格式
- Enhance throughput, lower memory bandwidth
- Reduce quantization error by “scaling”



參考資料: <https://developer.nvidia.cn/blog/introducing-nvfp4-for-efficient-and-accurate-low-precision-inference/>

NVFP4-like

- 為了讓同學更簡單地實作 NVFP4 ，採用 NVFP4-like (FP32 $\rightarrow$ bfloat16)
- 計算過程中不含 subnormal / NaN / Inf ，也就是 E 不會全為 0 或全為 1
- $x_{\text{real}} \cong (x_{\text{FP4}}) * (\text{factor}_{\text{FP8}}) * (\text{factor}_{\text{bfloat16}})$



NVFP4-like – 加法

1. 將 $\text{input}_{\text{FP4}}$ 換成 $\text{input}_{\text{bfloat16}}$
2. 將 $\text{factor}_{1, \text{FP8}}$ 換成 $\text{factor}_{1, \text{bfloat16}}$ ，並乘上 $\text{input}_{\text{bfloat16}}$
3. 再乘上 $\text{factor}_{2, \text{bfloat16}}$ 得到 $\text{input}_{\text{real, bfloat16}}$
4. 還原成 1.Fraction (8-bit)
5. 對齊小數點 (right shift 小的數，對齊大的數)
6. 對齊後把 significand 轉成二補數 (依 sign 決定是否取補數)
7. $\text{sum} = a + b$
 1. 判斷正負，可得 Sign
 2. 找到第一個 1 的位置，四捨五入，可得 Fraction
 3. 根據小數點移動多少，就把大的數的 Exponent 加減多少

NVFP4-like – 加法 example1

- FP4 of a = 0 01 1; FP4 of b = 0 11 1
- scale factor of a = 0 1001 010; scale factor of b = 0 0101 100
- scale factor of bfloat16 = 0 10000001 1010000

(real a) + (real b)

$$= [(1.1 * 2^{(1-1)}) * (1.01 * 2^{(9-7)}) + (1.1 * 2^{(3-1)}) * (1.1 * 2^{(5-7)})] * (1.101 * 2^{(129-127)})$$

$$= [1.111 * 2^2 + 10.01 * 2^0] * (1.101 * 2^2)$$

$$= [1.111 * 2^2 + 0.1001 * 2^2] * (1.101 * 2^2)$$

$$= (10.0111 * 2^2) * (1.101 * 2^2)$$

$$= 11.1111011 * 2^4$$

$$= 1.11111011 * 2^{(132-127)}$$

$$\rightarrow \textcolor{red}{1} \textcolor{green}{10000100} \textcolor{blue}{1111101}$$

NVFP4-like – 加法 example2

- FP4 of a = 0 01 1; FP4 of b = 0 11 1
- scale factor of a = 1 1001 010; scale factor of b = 0 0101 100
- scale factor of bfloat16 = 0 10000001 1010000

(real a) + (real b)

$$= [(1.1 * 2^{(1-1)}) * (-1.01 * 2^{(9-7)}) + (1.1 * 2^{(3-1)}) * (1.1 * 2^{(5-7)})] * (1.101 * 2^{(129-127)})$$

$$= [-1.111 * 2^2 + 10.01 * 2^0] * (1.101 * 2^2)$$

$$\rightarrow [10.001 * 2^2 + 0.1001 * 2^2] * (1.101 * 2^2)$$

加負數，先取補數

$$= (10.1011 * 2^2) * (1.101 * 2^2)$$

$$\rightarrow -1 * 1.0101 * 1.101 * 2^4$$

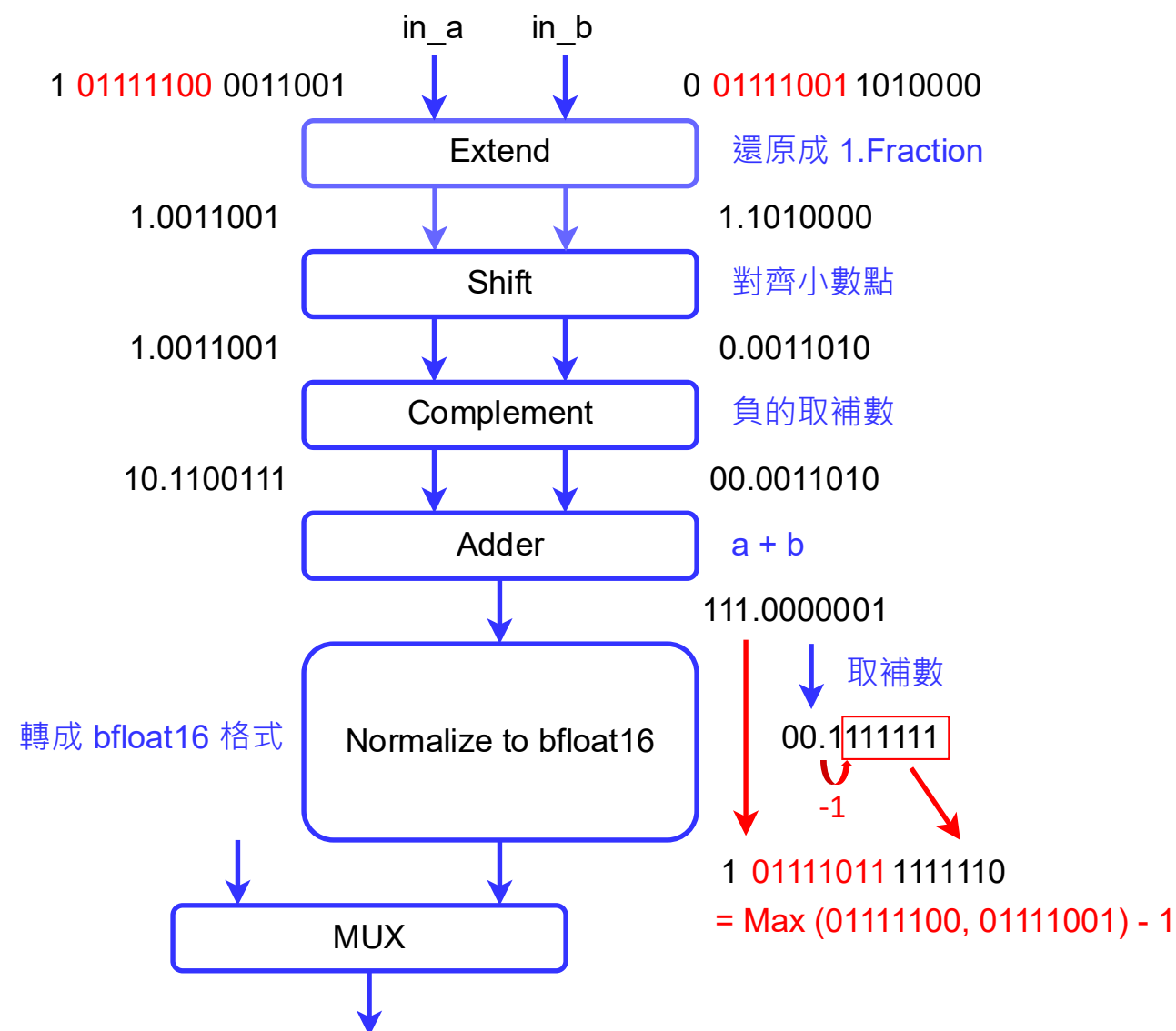
接下來要做乘法 → 取補數，-1 拿出來

$$= -1 * 10.0010001 * 2^2$$

$$= -1 * 1.00010001 * 2^{(130-127)}$$

$$\rightarrow 1 \ 10000010 \ 0001000$$

bloat16 加法 Diagram



NVFP4-like – 乘法

1. 將 $\text{input}_{\text{FP4}}$ 換成 $\text{input}_{\text{bfloat16}}$
2. 將 $\text{factor}_{1, \text{FP8}}$ 換成 $\text{factor}_{1, \text{bfloat16}}$ ，並乘上 $\text{input}_{\text{bfloat16}}$
3. 再乘上 $\text{factor}_{2, \text{bfloat16}}$ 得到 $\text{input}_{\text{real, bfloat16}}$
4. 還原成 1.Fraction (8-bit)
5. $a * b$
6. 轉成 bfloat16 格式
 1. 判斷正負($\text{sign}_a \text{ XOR } \text{sign}_b$)，可得 Sign
 2. 找到第一個 1 的位置，四捨五入，可得 Fraction
 3. $\text{Exponent} = \text{Exponent}_a + \text{Exponent}_b - 127$ (+n, if 小數點左移 n)

NVFP4-like – 乘法 example

- FP4 of a = 0 01 1; FP4 of b = 0 11 1
- scale factor of a = 0 1001 010; scale factor of b = 0 0101 100
- scale factor of bfloat16 = 0 10000001 1010000

(real a) * (real b)

$$= [(1.1 * 2^{(1-1)}) * (1.01 * 2^{(9-7)}) * (1.101 * 2^{(129-127)})] *$$

$$[(1.1 * 2^{(3-1)}) * (1.1 * 2^{(5-7)}) * (1.101 * 2^{(129-127)})]$$

$$= [1.111 * 2^2 * 10.01 * 2^0] * (1.101 * 2^2)^2$$

$$= [1.111 * 10.01 * 2^2] * (1.101 * 2^2)^2$$

$$= (100.00111 * 2^2) * (1.101 * 2^2)^2$$

$$= 1011.00100011111 * 2^6$$

$$= 1.01100100011111 * 2^{(136-127)}$$

$$\rightarrow 1 \text{ 10001000 } 0110010$$

bloat16 乘法 Diagram

